# Risk Management Framework

Carnivore Quant Fund, LP

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STT Capital Advisors, LLC

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## Risk Management Philosophy

The Carnivore Quant Fund is engineered for **capital preservation first, alpha generation second**. Every aspect of the system, from signal selection to position sizing to exit discipline, ensures the portfolio can absorb adverse conditions without meaningful drawdown. The Fund employs a multi-layered risk architecture spanning position sizing, portfolio-level controls, and automated alert systems, all enforced systematically with zero discretionary override.

## Position Sizing: 1% Vitality Cap

The Fund sizes full positions between **0.5% and 2.0% of NAV**, with **1.0% of NAV** as the standard allocation for individual equities and **0.5% of NAV** for ETFs. Because the initial entry (Lot 1) deploys only 35% of the full position, the actual capital at risk on any new trade is just **0.35% of NAV**, ensuring minimal impact from any single entry. Share count is calculated as floor(risk budget / risk per share). A wider stop produces fewer shares, never more risk.

| Asset Type | Max Position Size | Initial Entry (Lot 1) | Rationale |
|---|---|---|---|
| Individual Equities (Vitality) | 0.5%-2.0% of NAV (1.0% standard) | 0.35% of NAV (at 1.0%) | Core risk unit; full size earned through pyramid confirmation |
| ETFs | 0.5% of NAV | 0.175% of NAV | Reduced allocation reflects lower alpha potential |

## 5-Lot Pyramid System

Initial entry deploys only **35% of the full position**. Subsequent lots at +3%, +6%, +10%, +14% are earned through sequential confirmation, each lot requiring the prior lot to be filled, a time gate to be cleared, and a price trigger to be reached. Maximum capital is only deployed when the market has confirmed the trade multiple times.

| Lot | Name | Alloc | Trigger | Gate | Purpose |
|---|---|---|---|---|---|
| Lot 1 | The Scent | 35% | Signal entry | None | Initial position; market must confirm |
| Lot 2 | The Stalk | 25% | +3% from entry | 5 trading days | Largest add; time + price required |
| Lot 3 | The Strike | 20% | +6% from entry | Lot 2 filled | Momentum continuation confirmed |
| Lot 4 | The Jugular | 12% | +10% from entry | Lot 3 filled | Trend extension |
| Lot 5 | The Kill | 8% | +14% from entry | Lot 4 filled | Maximum conviction; full position |

## 10% Max Portfolio Risk Exposure

At no point does the Fund risk more than **10% of total portfolio NAV** across all open positions combined. This portfolio-level risk ceiling ensures that even in a worst-case scenario where every open position hits its stop

simultaneously, the maximum drawdown is capped at 10%. Recycled positions (stop beyond entry) carry $0 risk and do not count toward this limit.

## Portfolio Heat Caps

Open risk ("heat") is further segmented by direction to prevent overweighting either side of the book:

| Control | Cap | Description |
|---|---|---|
| Long Heat | 10% of NAV | Total open risk across all long positions |
| Short Heat | 5% of NAV | Total open risk across all short positions |
| Combined | 15% of NAV | Theoretical max if both sides fully deployed |

## Sector Concentration Limit

Net directional exposure is capped at **3 positions per sector**, calculated as the absolute difference between long and short positions within each sector. Prevents correlated drawdowns from sector-specific events. ETFs are exempt. When a sector breach is detected, the Risk Advisor provides two resolution options:

- **Option A:** Close the weakest position in the over-concentrated sector
- **Option B:** Add an opposite-direction position using top Kill candidates

## Stop Loss System: Proprietary PNTHR Stop Loss System (PPSLC)

All positions are protected by the **PPSLC**, a proprietary stop loss calculation that determines the most conservative of two variables and applies that price as the PNTHR Stop on all trades. The system allows price movement while remaining ready for counter-trend adverse directional risk. Stops never move against the trade:

- **BL positions:** Stop ratchets UP only (never moves down)
- **SS positions:** Stop ratchets DOWN only (never moves up)

### Stop Ratchet on Each Lot Fill

| Lot Fill Event | Stop Moves To | Effect |
|---|---|---|
| Lot 2 fills | Initial stop (unchanged) | Time + price confirmed, position monitored |
| Lot 3 fills | Average cost (breakeven) | Capital protected; initial investment covered |
| Lot 4 fills | Lot 2 fill price | Lot 2 gain locked in as minimum exit |
| Lot 5 fills | Lot 3 fill price | Full pyramid; aggressive ratcheted stop |

## Systematic Exit Discipline

Every exit is categorized and scored for discipline. Manual overrides are tracked and penalized. The system rewards systematic behavior:

| Exit Type | Trigger | Discipline Score |
|---|---|---|
| PNTHR Signal | Proprietary PNTHR Exit Signal is generated | 12/12 (Perfect) |
| FEAST | RSI > 85 momentum exhaustion, sell 50% immediately | 12/12 (Perfect) |
| PNTHR PPSLC Stop Hit | Ratchet stop hit | 10/12 |
| RISK_ADVISOR | Sector/portfolio concentration breach | 10/12 |
| STALE_HUNT | 20-day position without development, mandatory closure | 10/12 |
| MANUAL | Discretionary exit | 4/12 (profit) or 0/12 (loss) |

## Automated Alert Systems

### FEAST Alert (Momentum Exhaustion)

When weekly RSI exceeds 85 on any long position, the system triggers a **FEAST Alert**, a high-urgency notification to sell 50% of the position immediately. RSI > 85 historically signals extreme momentum exhaustion with high reversal probability.

### Stale Hunt Timer

Positions that fail to develop are automatically flagged based on trading days since entry:

| Trading Days | Status | Action |
|---|---|---|
| 15-17 days | STALE (Yellow) | Review position thesis |
| 18-19 days | STALE (Orange) | Prepare for liquidation |
| 20+ days | LIQUIDATE (Red) | Mandatory position closure |

## Wash Sale Compliance

30-day re-entry lockout on losing trades, **automatically enforced by the pipeline**. Any attempt to re-enter a position within 30 calendar days of closing at a loss is blocked at the pre-trade scoring level (Analyze Score penalizes wash sale violations with 0/5 points).

## Pre-Trade Risk Assessment (Analyze Score)

Every potential trade is scored through the **Analyze system**, a 100-point pre-trade assessment where every point is evaluable at scan time. No estimation, no guesswork:

| Tier | Weight | Components |
|---|---|---|
| T1: Setup Quality | 40 points | Signal Quality (15), Kill Context (10), Index Trend (8), Sector Trend (7) |
| T2: Risk Profile | 35 points | Freshness (12), Risk/Reward (8), Prey Presence (8), Conviction (7) |
| T3: Entry Conditions | 25 points | Slope Strength (5), Sector Concentration (5), Wash Compliance (5), Volatility/RSI (5), Portfolio Fit (5) |

Score >=75% = green (optimal). >=55% = yellow (proceed with awareness). <55% = red (reconsider). SIZE IT is blocked when Analyze detects errors in underlying data.

## Macro & Sector Gates

The orders pipeline applies additional gates before any position can be opened:

- **Macro Gate:** SPY/QQQ regime assessment (21-week EMA direction and multiplier)
- **Sector Gate:** Sector EMA direction must align with trade direction
- **D2 Gate:** Kill scoring dimension 2 (sector direction) must not be negative
- **SS Crash Gate:** SPY/QQQ EMA falling for 2 consecutive weeks AND sector 5-day momentum below -3%

## Worst-Case Validation (MAE Analysis)

The maximum adverse excursion (MAE) across all 2,520 closed pyramid trades was **-15.2%** on a single trade. At Lot 1 sizing (35% of the full position = $3,500 on a $10K position), this translated to approximately **0.5% of portfolio NAV**.

Even during the months when worst-case MAE trades occurred, the portfolio remained profitable on a net basis. The 1% vitality cap and 35% initial lot sizing ensure that no single adverse trade can materially impair investor capital. No drawdown resulted in permanent capital loss.